

What is an API?

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API stands for Application Programming Interface.

It allows two applications or systems to communicate with each other.

Simple Example

When you use a weather app:

1. The app sends a request to the weather server using an API.
2. The server sends weather data back.
3. The app shows the result to the user.

So API works like a messenger between applications.

Real-Life Example

Frontend (React website)

Backend (Flask/Python server)

The frontend sends a request:

GET /users

Backend returns data:

```
{  
  
  "name": "Ali",  
  
  "age": 22  
}
```

Common API Methods

GET → Fetch data

POST → Send/Create data

PUT → Update data

DELETE → Remove data

Example in Python Flask

```
from flask import Flask
```

```
app = Flask(__name__)
```

```
@app.route('/hello')
```

```
def hello():
```

```
    return "Hello API"
```

```
app.run()
```

When user opens:

`http://127.0.0.1:5000/hello`

API returns:

Hello API

Types of APIs

REST API → Most common web API

SOAP API → XML-based API

GraphQL API → Flexible querying API

Why APIs are Important?

- Connect frontend and backend
- Integrate third-party services
- Enable automation
- Share data securely
- Used in mobile and web apps

Examples of APIs

- OpenAI API
- Google Maps API
- GitHub API
- Weather API by OpenWeather